

Yuyang Hu — Consolidated Knowledge Base

*Consolidated from the Pinecone [memories](#) index (306 records across 17 source documents).
Chunk IDs and timestamps stripped; scattered chunks stitched back together by subject;
duplicates merged. Generated 2026-05-30 from the raw inventory dump.*

1. Identity & Bio

- **Full name:** Yuyang Hu. Born **October 29, 2006** in Beijing, China (Haidian district), to **Jianling Wang** (mother) and **Yuming "Steve" Hu** (father).
- **Also known as "Beric"** by friends. Went by "Eric" in elementary school; friends added the "B" (for "Big") to distinguish him from Eric Tran ("Eonx"), the other Eric in the group.
- **Height:** 5'11".
- **Hometown:** San Diego, CA. Moved to the US Nov 4, 2011 (age 5); grew up in Torrey Highlands, North County San Diego.
- **Schools:** Deer Canyon Elementary → Mesa Verde Middle School → Westview High School → NYU.
- **University:** NYU, sophomore, dual major in **Mathematics & Computer Science** at the Courant Institute. Expected graduation **May 2028**. Based in Union Square, NYC.
- **Roles:** VP at **Tech@NYU** (promoted April 2026; previously Mentorship Co-Director/Lead). **Director of the ML team at the Stern Business Analytics Club (BAC)** — Spring 2026. Interning at **MicroAGI** in a growth/expansion role on **Shift**.
- **Contact:** yh5525@nyu.edu · 858-428-4832.

Interests & personal context

- Grew up between Beijing and the US. Heavy **science-fiction** reader: Asimov's *Foundation* and Robot/Daneel Olivaw series; Frank Herbert's *Dune* (read all of them, favorite is *God Emperor of Dune*).
- **Grand strategy & games:** Stellaris, Hearts of Iron, Minecraft, Among Us, Rainbow Six Siege.
- **History & geopolitics:** Chinese history, Middle Eastern politics, Southeast Asian demographics; developed a love of history through map games.
- **Active:** runs regularly (≥2 miles daily), hiking, exploring NYC on foot.
- **Music:** plays acoustic guitar, fond of country music (favorite country song: **"Springsteen" by Luke Combs**); does lyric analysis of Zach Bryan and Sons of the East.
- Also: video games, history documentaries, museums, culinary exploration, learning **German**.

- Uses Claude as both a technical collaborator and a reflective sounding board / grounding mechanism.
- Misc likes: the beach, the ocean, apples.

Working preferences

- Render math in **LaTeX**.
 - Prefers very basic explanations when first introducing a concept, scaling up in depth as needed.
 - Prefers structured, copy-paste-ready outputs — markdown, single-block formats for reuse.
 - Wants clear architecture explanations when designing ML systems, pipelines, or backend infra; implementation-oriented responses with concrete code and step-by-step plans.
 - Frequently writes `CLAUDE.md` / `copilot-instructions.md` and structured config docs.
 - Wants a remembered, ongoing **step-by-step roadmap for entering AI network/performance optimization on hardware** (skills + projects).
 - Working on an app for TestFlight / App Store beta distribution.
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2. Life Timeline (Core Memories)

- **Oct 29, 2006** — Born in Beijing (Haidian). First five years lived in Beijing (two-story apartment in Beiyuan, north Beijing).
 - Early childhood: Lego classes; family had a red 2003 Citroën Élysée; roller skating; maternal grandfather owned a chestnut farm in Pinggu (weekend visits, farm animals); visited the Great Wall once; very close with a cousin (father's sister's daughter); grandmother dropped him at preschool.
- **2008** — Parents took him to the Beijing Summer Paralympics (first major sporting event); family lived near the Olympic park.
- **Nov 4, 2011** — Moved to the US; he and his mom came to San Diego ~4 months after his dad. First meal in America was (ironically) Chinese food.
- **Dec 2011** — Started kindergarten at Sage Canyon Elementary; struggled with English, attended ESL classes.
- **Aug 2012** — Family moved to their current house in Torrey Highlands; started Deer Canyon Elementary.
 - General US childhood: two cars (silver 2011 Honda Odyssey, white 2011 Honda CRV); house nearly burned in a 2014 wildfire; favorite beach Torrey Pines (weekend hikes after dinner); returned to China almost yearly until COVID; road trips across North America (LA, SF, Portland, Seattle, Salt Lake City, Phoenix, Grand Canyon, Yellowstone, Bryce, Zion). **Michael Wang** (transferred from

China in 3rd grade) was a best friend in elementary school; families dined together; Canada trip in middle school.

- **April 2017** — Completed the **Ride Across California** with his dad (Yuma → San Diego), a 5th-grade tradition.
- **June 2017** — Graduated Deer Canyon; began Mesa Verde Middle School.
- **2018** — Met **James "Jimmy" Buchmiller** (future best friend) and **Sutchin Somanathan** (closer with Sutchin at the time; lots of Minecraft in 6th/7th grade).
- **2019** — Grew distant from Sutchin, closer to Jimmy's friend group (Jimmy, Tim Waters, Ian Papworth, Grant Foster, Nate Thomas); met **Keralily Smith**.
- **2020** — COVID, online school, video games (Among Us, Hearts of Iron, Minecraft, R6 Siege); picked up guitar; developed a love of history via map games.
- **2021** — Adopted family dog **Chester** (German Shepherd, <2 months old) from a farm in Temecula; began Westview High; 8th-grade graduation held virtually; summer camping in the Mt. Cuyamaca mountains; returned to in-person school.
- **2022** — Eventful: first real relationship, AP classes, school events.
- **2023** — Started thinking about college; wanted to live in NYC → applied early decision to **NYU CAS for Math & Econ**.
- **2024** — Graduated HS; returned to China for the first time in ~5 years; moved to NYC in August; rough adjustment to the competitive environment; began studying hard, took more interest in CS. **Nov 2024** placed 2nd at the NYU DS club project expo (first CS competition); switched major to Math & CS.
- **2025** — Fully immersed at NYU; joined the **BAC Stern ML team**; first research project **PAINT** (diffusion art restoration); ran a summer camp teaching HS students Arduino/CAD/AI; grew interested in ML architecture & performance optimization. **Late summer**: built first iOS app, **Journey**. **Fall**: became PM for BAC ML (led **dAlgnosis**), joined Tech@NYU Mentorship; 2nd place at **NYU Archaeohack** (Egyptian hieroglyph classification).
- **2026** — Joined eboard for both BAC and Tech@NYU (ML lead and mentorship lead, respectively); more projects: **Nootes**, **Cronicl**, and implementing his own theoretical architectures (e.g. in Journey).

3. People

Merged from the contact list, one-off notes, and recent updates. Recent updates supersede older descriptions where they conflict.

Family

- **Steve (Yuming) Hu** — Dad. Works in ML at Apple. Big source of inspiration. BS in Electrical Engineering (Beijing Institute of Technology), MS in Data Science (UCSD). Watching his master's projects in middle school first sparked Yuyang's interest in ML.

- **Jianling Wang** — Mom. Strict growing up (drove a strong work ethic). BS and MS in CS from the Beijing Institute of Technology; worked at startups before the family moved to the US.
- **Sally Hu** — Little sister, 7 years younger, a fantastic dancer, speaks better Chinese than Yuyang. (Note: "Yuying" is a hallucinated/incorrect name — her name is Sally.)

Friends & network (alphabetical)

- **Md Abedlin** — Tech@NYU mentor, friend of Jean Park; recently had dinner; mentoring someone named Jerry Hu.
- **Amaan Arif** — Freshman-year roommate; rooming together again next year. Had a class with Barron Trump at NYU Stern freshman year (group lore: tried to get Amaan to invite Barron to their Minecraft server; gifted Yuyang a parody Dune poster of him and Barron).
- **Avinash** — Lived on the same floor freshman year.
- **Anton Baranov** — Chinese school together in elementary; grew up together.
- **Rayan Basheer** — BAC ML, Fall 2025.
- **Vivek Bellum** — BAC eboard member.
- **Nate Blum** — Met in Lipton freshman year; went to a concert together in 2025.
- **Brandon Boswell** — Played Hearts of Iron together; trips together.
- **Erron Bridgewater** — Tech@NYU mentorship team member.
- **James "Jimmy" Buchmiller** — One of his closest friends from back home; met through Michael Wang in middle school; same friend group since; many trips together.
- **Angel Checo** — Sophomore-year suitemate at NYU.
- **Aidan Chen** — High school together; volunteered together with Tzu Chi.
- **Sophia Chen** — US Arts academy (San Diego) in middle school, same high school, both now at NYU.
- **Sienna Chan** — Suitemates with Makaila; knows Ethan and Martin; did a Datathon together.
- **Joseph Cheng** — BAC ML Spring 2025 cohort; current Quant team co-director; chill.
- **Freddy Chieng** — Grew up together.
- **Bostyn Connelly** — Ex-girlfriend (dated a year in senior year HS); still in touch; one of the most encouraging, supportive people he knows.
- **Danielle Copeland** — Taking CS classes together at NYU since freshman fall; never shows up to class.
- **Pope Cruz** — Tech@NYU eboard; chill; brought Jollibee to the first eboard social.
- **Cyan** — Tech@NYU eboard; chill.
- **Dan Dan** — Boss at Haddee; hard worker.
- **Daniel Dao** — AP CSA together sophomore year HS.
- **Nicholas Didanato** — Grew up together.
- **Kevin Dong** — BAC and Tech@NYU eboard; many hackathons together (Tech@NYU Buildathon, Gemini NYC); great at UI design.
- **Kaikai Du** — Tech@NYU and Gemini hackathon teammate; smart; ML + fullstack.
- **Gabe Estrov** — Closest friend from back home; met in HS; house in Santa Cruz (visited twice); play CS:GO together.

- **Donnie Feng** — Grew up together, NYU together; BAC Insight + Tech@NYU Tech Treks member.
- **Grant Foster** — Closest friend from back home; middle + HS; many lake-house trips; annual game of 8-ball (one turn each per year).
- **Raymond Fu** — BAC ML S26 cohort.
- **Anish Gowda** — Classes together every year from 6th grade through HS graduation; met during the Ride Across California in 5th grade.
- **Echo Shi Gu** — Advisor for NYU's Tzu Ching chapter; helped him adjust to NYC freshman year.
- **Jocelyn Gu** — In his PM group for BAC ML.
- **Ryan Guan** — Tech@NYU Buildathon + Gemini Hackathon; let the group use his apartment; pulled two all-nighters building Nootes.
- **Saloni Gupta** — Bigger Bobst warrior than Yuyang.
- **Deven Huang** — BAC ML when Yuyang was PM; came up with the dAlgnosis idea.
- **Jennifer Huang** — Tech@NYU eboard member.
- **Marko Huisa** — High school together; goes to ASU; wants to talk more.
- **Siya Jain** — BAC VP.
- **Rahul Jayachandran** — BAC President; big Quant guy; very smart.
- **Braiden Johnston** — AP European History together sophomore year HS.
- **Jae Jeon Jung** — Grew up together; NYU Tandon; loves biking and photography.
- **Kevin Kakkad** — Met the summer after freshman year; went to the same HS (didn't know each other then); BAC ML team member.
- **Makaila Kim** — NYU Archaeohack teammate. **Update: "not mid" — she's not just okay anymore.**
- **Pranay Kulkarni** — HS together, college together; met in AP Physics senior year HS.
- **Sean Lai** — President of Tech@NYU; cracked at networking.
- **Hugo Lee** — Data Science club Datathon together freshman year; hard worker.
- **Owen Leung** — PM for BAC ML S26; fun to hang out with. **Update: new ML Director for BAC ML (F26).**
- **Adam Levine** — Grew up together; one of the smartest guys he knows.
- **Ray Levine** — HS together, AP CSA sophomore year; Airbnb trip in Julian, CA over freshman winter break.
- **Sarina Li** — BAC eboard member.
- **Daniel Liang** — BAC eboard; big runner; always likes his Strava activities.
- **Megan Lin** — Linear Algebra together freshman year.
- **Coco Liu** — Tech@NYU eboard member.
- **Max Liu** — Goes to Dartmouth; one of the smartest people he knows; grew up together; research project together summer 2025.
- **Reynold Liu** — Saw each other a lot at the Lipton dining halls freshman year.
- **Sam Luo** — Both live in Carlyle; Australian; cool to hang out with.
- **Malek Maaz** — Grew up together; one of the few to visit him in New York.
- **Ethan Madrideo** — High school together.
- **Vayun Malik** — Co-director for BAC ML; one of the smartest people he knows; no spatial awareness.

- **Haley Ngai** — Tech@NYU eboard member.
- **Michael Odekunle** — Good friend at NYU; also a Math + CS major; in his PM group for BAC ML.
- **Jean Park** — Co-director for BAC ML; very hard worker. **Update: now working at Palantir.**
- **Ian Papworth** — Closest friend from HS; first to rep Super Joey San Diego merch; big chiller.
- **Ryland Pimentel** — His "best bro"; grew up together; call most weeks; got him through rough times in college.
- **Ritvik Rallapalli** — His director for two semesters of BAC ML.
- **Ashish Reddy** — BAC ML, Fall 2025.
- **Mia Rodriguez** — Tim Waters' girlfriend; close friend; loves her dog Pom Pom.
- **Colin Roland** — Met in middle school; closest friend from back home; visited NYC over Thanksgiving 2025 (one day's warning); drove down from Santa Cruz together summer 2025.
- **Eshaan Saxena** — His PM in BAC ML, then his director when he was a PM; a "fatherly" figure, great for advice.
- **Emily Silkina** — Met through BAC freshman spring; smart, hard worker.
- **Ajit Sivakumar** — Went through BAC ML together; PM for BAC ML S26; good fencer; into history, running, and plays Yuyang's guitar.
- **Keralily ("Kera") Smith** — Closest friend from HS; hard worker; admires her progress.
- **Larson Smith** — "Lardog"; chill to talk to.
- **Sutchin Somanathan** — Met in middle school; best friends in 7th/8th grade; most Zen person he knows.
- **Nihal Sundari** — Met during welcome week freshman year; always has the best fits.
- **Francis Tian** — Grew up together.
- **Nate Thomas** — Closest friend from back home; loves his dog; Tahoe lake-house trip summer after freshman year.
- **Eric Tran ("Eonx")** — Middle + HS together; admires his singing and filmmaking.
- **Kuuga Ueda** — Met during welcome week freshman year; still gets dinner together; nice and hardworking.
- **Chengzhi Wang** — Long story how they met (ask in person); super smart and nice.
- **Martin Wang** — Ethan Xu's old freshman-year suitemate (Lipton); artistic; data science major; NYU DS club & Marketing Society.
- **Tim Waters** — One of his best friends from back home; met in middle school; goes to Cal Poly Pomona; wants to be a Disney Imagineer; signatory to the founding of the "Sovereign Republic of Yezustan." Wrote the song "Winner" sophomore year HS (later Christmas/2025 re-releases).
- **Matt (Matthew) Wilson** — Crazy nice; took ~half the photos on the portfolio site; talented filmmaker/photographer; film student transferring into SDSU.
- **Angelina Wu** — Met during welcome week freshman year; fun.
- **Ethan Xu** — Crazy smart; part of his group when he was a PM for BAC ML.
- **Luke Xu** — Met online before college; one of the people he's known longest at NYU.

- **Ray Yang** — One of his closest friends freshman year; showed him around Shanghai the summer after freshman year.
- **Charles Zhang** — Met in elementary school; very smart.
- **Isaac Zhang** — Met during welcome week freshman year; a "day one" at NYU.
- **Forrest Zheng** — In his group when he went through BAC ML.

Lore & inside jokes

- **"Beric"** — his nickname (see Bio).
- **"Charging the battery"** — a phrase that invokes deep sorrow; apologize profusely if it comes up, then move on.
- **"Snake in the brakes"** — from camping near Mt. Palomar the summer after junior year; friends teased him over his fear that his car's brakes would fail combined with his fear of snakes.
- **"Chester feypan"** — dog Chester responds positively to "feypan" (a butchered spelling of *feipan*, pinyin for frisbee); Kera loves saying it.
- **Nate Thomas** pooped his pants in Yuyang's car while camping at Mt. Palomar (summer after junior year).
- **Matt Wilson** got high while camping spring break freshman year, got freaked out by the campfire, and yelled "SILENCE!!!!" whenever people talked.

4. Skills & Resume

Summary profile: Sophomore at NYU studying CS & Math, expected to graduate May 2028. Technically versatile developer with hands-on experience in full-stack web dev, mobile (Swift/iOS), ML, and AI systems. Most sophisticated work: on-device LLM fine-tuning with Apple CoreML, diffusion-based art restoration, and RAG pipelines for hallucination reduction in medical contexts. Active student leader (co-director of BAC ML, mentorship lead at Tech@NYU). Two 2nd-place finishes in NYU competitions (DS Project Expo, Archaeohack).

Education: NYU — BA in Computer Science & Mathematics, expected May 2028. Relevant coursework: Basic Algorithms, Data Structures, Computer Systems Organization, Numerical Analysis. Awards: 2nd Place NYU DS Project Expo; 2nd Place NYU Archaeohack.

Technical skills:

- *Languages:* Python, C, C++, Swift, Java, JavaScript, SQL, HTML, CSS.
- *Frontend:* React, HTML, CSS.
- *Backend:* Node.js, Spring Boot, Django, NestJS.
- *Databases:* MongoDB, PostgreSQL, SQL, Supabase, Pinecone.
- *DevOps & cloud:* Docker, Git, AWS, Railway, Vercel.
- *ML & AI:* CUDA, PyTorch, TensorFlow, CoreML, Hugging Face, diffusion models, LLMs, RAG pipelines.
- *APIs:* OpenAI API, YouTube Data API v3, REST APIs.

- *Other (self-listed)*: mentorship, project management, event planning, technical interview prep, data scraping, professional communication, video production/editing, business marketing, social media, leadership.

Work experience:

- **Flerish — Campus Events Intern** (June 2025 – present, NYC): coordinated logistics and PR for activations of 300–700 attendees; partnered with Gunna's tour team on NYC product-promotion campaigns (500+ direct consumer interactions).
- **Haddee Education — Software Engineering Intern** (June 2021 – Aug 2023, San Diego): built a full YouTube Data API v3 integration to dynamically embed video content on the company site; cut manual content-update time ~70%; built a REST API pipeline for video assets/metadata.

Leadership:

- **NYU Stern Business Analytics Club — Co-Director, ML Team** (Jan 2025 – present): led a 15-student cohort through neural nets, CNNs, transformers (100% completion); directed a semester project fine-tuning open-source LLMs on synthetic medical datasets with CoT + RAG, reducing hallucination ~35%.
- **Tech@NYU — Co-Director, Mentorship Team** (Sept 2025 – present): guided 20+ students through mentor-mentee pairings with NYU alumni; recruited 5+ industry professionals for technical panels.

5. Research Interests

- **Efficient Model Architectures** — small-parameter and quantized networks that stay competitive under compute constraints; inference optimization for resource-limited environments.
 - **Computational Behavioral Modeling** — ML frameworks to simulate human behavior, neurological activity patterns, and large-scale agent dynamics, including modeling nation-states as autonomous agents within geopolitical world models.
 - **Alignment & Fine-Tuning Methodologies** — RL-based training paradigms (RLHF, GRPO) for agent alignment, robustness, and task-specific performance in language and embodied settings.
 - **Retrieval-Augmented & Adaptive Systems** — RAG for reasoning/knowledge grounding; on-demand model training within procedurally generated or custom 3D simulation environments.
 - **Language & Linguistic Applications** — semantic modeling, language structure analysis, cross-lingual generalization.
 - **Full-Stack ML Integration** — end-to-end systems bridging research-level ML with production React apps for real-time interaction and deployment at scale.
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6. Projects

GitHub repository overview (@yuyanghu06)

Repo	Lang	Updated	Summary
yuyang-site	TypeScript	Mar 5, 2026	Personal portfolio site (React/TS). Central hub with a fine-tuned LLM that mimics his speech, augmented with an MCP layer of ground-truth memories.
journey	Swift	Mar 4, 2026	Native iOS journaling app; on-device CoreML personality model surfacing Big Five insights; Swift + NestJS + PostgreSQL; privacy-first, local fine-tuning.
cronicl	TypeScript	Mar 2, 2026	Gemini Hackathon (Spring 2026) chronicle/logging app using the Gemini AI API.
CSCI-UA-473 Fundamentals of ML (S26)	Python (MIT)	Jan 21, 2026	Fork of the official NYU course repo (Kyunghyun Cho); coursework/study.
yuyangGPT	Jupyter	Jan 1, 2026	Small-parameter LM trained on Yuyang's texting patterns using an HRM (Hierarchical Reasoning Model) architecture; explores HRMs for hyperspecific narrow roles to cut compute.
dAlignosis	Jupyter (1★)	Dec 7, 2025	Medical-advice LLM from BAC ML Fall 2025; prompt

			engineering, fine-tuning, RAG for a sensitive domain.
GreenTeamArchaeo Hack	Jupyter	Nov 16, 2025	NYU Archaeohack (Green Team); ML to classify Egyptian hieroglyphs; runner-up.
manifestLearningWebsite	HTML	Oct 8, 2025	Public site for Manifest Learning, an edtech org he contributed to.
Kaiwa-Site	HTML/JS	Mar 17, 2025	Kaiwa (会話) language-learning platform; toggleable real-time translation connecting learners with native speakers.
meanreversiontradingstrategy	C++	Feb 9, 2025	Mean-reversion trading strategy; Makefile build; signal generation, backtesting, execution.
meanreversionlinearregression	—	Oct 30, 2024	Mean reversion + linear regression; a more statistically rigorous evolution of the above.
presidentialSpeech Analysis	Java (1★/2 forks)	Dec 4, 2024	Analyzes linguistic complexity of presidential speeches via the Gunning Fog Index; NLP + computational linguistics + poli-sci.
nootes (collab @ka1kqi/nootes)	TS/Python/PLpgSQL (2★, Apache-2.0)	—	"Merge ideas, not just text." Collaborative AI document platform: share & fork docs, AI-powered merging, document generation, mind graphs, real-time collab.

			Stack: Vite+TS, FastAPI, PostgreSQL via Supabase, in-house GPT agent, Railway. Live: nootes-beryl.vercel.app
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Cuyamaca — natural-language control for Arduino robotics

Tauri v2 desktop app (macOS/Windows). Describe hardware in a visual **parts editor** (a *manifest*: board type, serial port, every component with pin assignments). A **code model** generates an Arduino sketch from the manifest, shows it for review, and flashes it with one click. After flashing, a **runtime window** opens: type natural-language commands ("move forward slowly", "scan for obstacles"); a **runtime model** reads live sensor data and camera feeds, decides actions, and sends serial commands automatically until you hit the kill button. Two independent model slots (code + runtime), each configurable (cloud API and/or local).

Sierra — local natural-language smart-home controller

Tauri v2 (Rust backend + React/TypeScript frontend). Connects to **Home Assistant** and controls devices by chatting with a local LLM via **Ollama**. No cloud, subscriptions, or accounts by default; optional external APIs (OpenAI/Anthropic/Google/Mistral).

- **Requirements:** macOS/Windows (Linux unsupported). Local LLM needs resources — 7B/13B model + 16 GB RAM recommended; tool-calling models work best.
- **Dependencies:** Ollama (port 11434) and Home Assistant Core (Python 3.12+, port 8123). Auto-detected and set up on first launch; can be managed manually.
- **Architecture:** React frontend → Tauri IPC → Rust backend (Ollama service with streaming chat + tool-calling loop; HA REST + WebSocket clients; Tool Registry — 19 tools across 4 domains; MCP server over HTTP port 3001; in-memory device state cache; process manager; installer; config store as 0600 JSON). Managed services: Ollama + Home Assistant.
- **Supported HA domains:** Lights, Switches, Climate, Media Players (more planned: locks, covers, fans, vacuums, cameras, alarm panels, scenes, scripts, automations — each just needs a new tool definition file).
- **Troubleshooting:** detects externally-running Ollama/HA on busy ports; HA onboarding flow; model-download failures; auto-restarts crashed services up to 3×; logs in [~/Library/Logs/Sierra/](#) ([ollama.log](#), [homeassistant.log](#)).
- Open source. Ollama: MIT. Home Assistant: Apache 2.0.

Journey (iOS app)

Reflective AI-powered journaling companion. SwiftUI + SwiftData, MVVM with protocol-based abstraction. **56 Swift files, ~12K LOC, 4 bundled ML models. No external dependencies** (no CocoaPods/SPM). Updated Mar 7, 2026.

- **Features:** chat-based daily reflection; AI journal generation; on-device personality model (inference + training); persistent local storage (SwiftData); Keychain auth with refresh cycle; calendar grid with data viz; memory/context document import; background task scheduling; calm, reflective design system.
- **4 tabs:** Today (ChatView) · Past Self (Personality) · Explore (Calendar) · Settings.
- **On-device PersonalityModel:** two-model architecture — a frozen backbone (`PersonalityModelStock`: 384-dim embedding → 512-dim latent → 325-dim logits) plus a trainable projection head (`PersonalityHeadUpdatable`: 512 → 325). Inference: embed → latent → logits → top-8 personality tokens (e.g. "reflective", "curious"); 325-token vocabulary. Auto-trains every 14 days via `BGProcessingTask` on the last 14 days of conversation + memory documents (head weights only; no backprop through the backbone). Embeddings via `MiniLMEmbedder` (all-MiniLM-L6-v2, 384-dim; falls back to `NLEmbedding`). All weights stay on-device; only tokens are sent to the backend for context.
- **Services:** `AuthService` (Keychain tokens, auto-refresh every 14 min), `ChatService`, `PersonalityModelService` (Actor), `ContextDocumentService` (PDF/text import), `PersonalityTrainingScheduler`.
- **Persistence:** SwiftData models (`MessageRecord`, `PersonalitySessionRecord`, `PersonalityMessageRecord`, `ContextDocument`); repositories are protocol-backed (in-memory + SwiftData implementations); `JournalRepository` is currently in-memory only.
- **Design system (DS namespace):** warm-neutral backgrounds, pastel accents (sage, dustyBlue, warmYellow, softLavender, blush), SF Pro Rounded, spring/easeInOut animations.
- **Requirements:** Xcode 15+, iOS 15+, M1/M2 Mac.
- **Roadmap:** SwiftData-backed journal repo; unit tests for ViewModels/services; real `BugReportService`; offline-first sync; multi-model personality; conversation search/filter; PDF export; push notifications for training completion; iCloud/multi-device sync; rich media; Apple Health integration.

Journey Backend

Server-side API for Journey. **NestJS + PostgreSQL (Prisma) + OpenAI (GPT-4o mini)**, deployed on Railway via Docker. Powers daily AI conversations, reflective journal generation, and a calendar of past activity. All AI calls are confined to a single service — the iOS app never calls OpenAI directly. Every piece of data is scoped to a **DayKey** (`YYYY-MM-DD`, user's local timezone).

- **API:** Auth (register/login/logout/refresh) · Chat (`GET /days/:dayKey`, `POST /chat/sendMessage`) · Journal (`POST /journal/generate`, upserts one entry per day) · Calendar (month activity indicators) · Days · Memories (CRUD; ≤200-char label, ≤50k-char content) · Personality (stateless, token-aware chat) · Health Check.
- **DB schema:** User, Message, JournalEntry (unique per user/day), RefreshToken (30-day TTL), Memory.
- **AI config:** OpenAI; 512 max tokens/response; temp 0.7; 24,000-char max context (oldest dropped first); retry on 429 and 5xx. User messages are persisted *before* the AI call, so they're never lost on failure.
- Railway deploy auto-runs migrations on each push; tests cover DayKey validation, persistence, journal upsert/idempotency, and AI-failure resilience.

Portfolio / Personal Site (yuyang-site)

Full-stack: **NestJS** (server) + **React/Vite** (client) served from a single Node process.

Photography-first editorial hero layout, an AI chatbot powered by **RAG over a personal knowledge base in Pinecone**, and a contact form delivering email via SMTP. Agentic features help users navigate the site and projects. The AI is a custom **LoRA agent on Qwen 3 235B**, fine-tuned on Yuyang's personal messages and documents.

- **Stack:** NestJS 10 / TS / Node ≥20; React 18 / Vite 5 / Router v6; TogetherAI (chat); OpenAI (embeddings); Pinecone (RAG); Nodemailer over SMTP (Gmail); Railway.
- **Agentic tools:** `[navigate]`, `[contact]`, `[redirect]` (map in `chatActions.ts`), `[message]` (default).
- **Knowledge base ingestion:** a script reads markdown source docs, splits into ~300-token overlapping chunks, embeds each via OpenAI, and upserts vectors + metadata into Pinecone (idempotent — overwrites by ID). If the RAG index is absent, chat falls back to system-prompt-only mode.
- **Dev:** Vite client on 5173, NestJS API on 3000 (Vite proxies to NestJS). Production build serves the React app as static files from the single Node process.

yuyangGPT — personal training pipeline

Pretraining + fine-tuning **Qwen 3.5-4B** on custom datasets (iMessages, documents, structured conversations). LoRA/QLoRA optimization, encrypted iPhone-backup support, multi-GPU training.

- **Data scripts:** iMessage scraper → JSONL (from the live macOS Messages DB or an iPhone backup, encrypted or not; only exports messages he sent); dataset converter (JSON/CSV → user-assistant conversation format); text extractor (PDF/DOCX/MD/TXT → JSONL chunks); newline-flattener; train/val splitter.
- **Training:** YAML-config driven. *Pretrain* (causal LM on raw text — iMessages, document dumps). *Fine-tune* (LoRA/QLoRA on formatted conversations). Resume from checkpoint; optional streaming datasets; Weights & Biases logging.

- **Inference & deploy:** interactive chat with a system prompt; merge LoRA adapter into base weights to produce a standalone model; optionally push to Hugging Face Hub.
- **Utilities:** dataset validation, token counting, model testing, sample generation.

Data Embedding Ingest Pipeline

Local pipeline that extracts text from PDF/Markdown/plain text, chunks it, embeds it with OpenAI, and upserts into Pinecone. Python 3.11+, a Pinecone index (1536-dim), and an OpenAI key. Web frontend with three ingest modes (Upload PDF, Upload Markdown, Paste Text) plus matching API endpoints.

PAINT

Art-restoration tool leveraging diffusion models, built as a research aid for academic collaborators. Curated a 50,000+ artwork dataset to train artwork-movement classifiers and image-reconstruction models; used a diffusion architecture for high-quality reconstruction/restoration; deployed via a scalable Flask backend. Tech: Hugging Face, PyTorch, NumPy, Flask, diffusion models.

Buildathon Project

(Appears to be a boilerplate/template README — generic "platform for collaborative learning, documentation, and project management" with placeholder install/usage/contributing sections. Likely a scaffold rather than a real shipped project.)

7. Quick facts

- **Hometown:** San Diego, CA.
- **Height:** 5'11".
- **Favorite country song:** "Springsteen" by Luke Combs.
- **BAC ML Director**, Spring 2026.
- **Jean Park** → now at Palantir.
- **Owen Leung** → new BAC ML Director (F26).
- Sister's correct name is **Sally** (not "Yuying").